

A

B

C

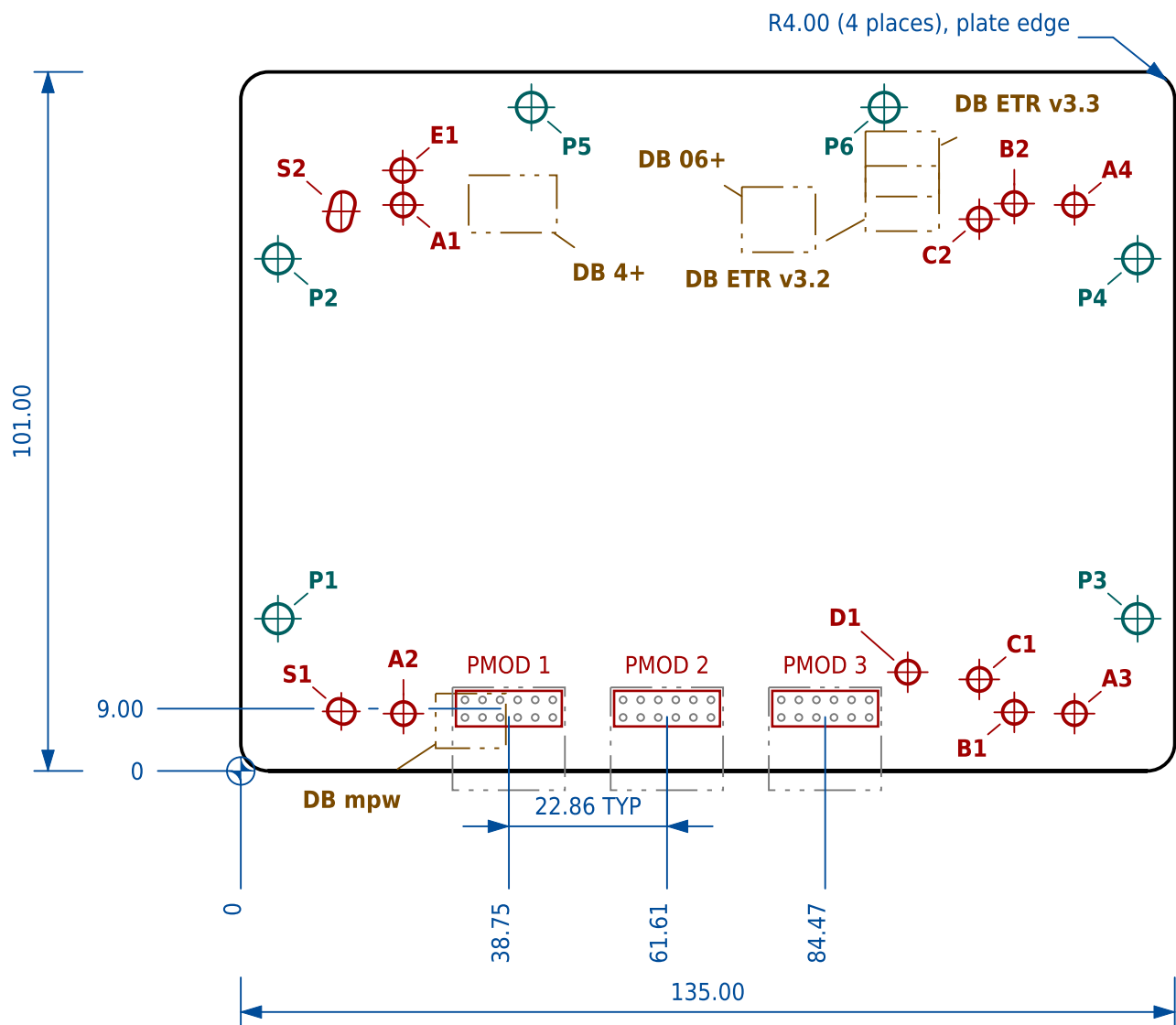
D

A

B

C

D



NOTES

- Viewed from the side the board mounts on.
- A slot row gives its two end centres; DIA/WIDTH is the slot width and LENGTH its overall length.
- USED BY names the board a feature serves and that board's own hole ID, MT1 to MT4. Drawing TT-MP-FIT shows each board on the plate.
- A hole's ID letter is the board it is there for, A DB mpw, B DB 4+, C DB 06+, D DB ETR v3.2, E DB ETR v3.3, numbered within it. S is a slot, P a plate fixing. A feature two boards share keeps one ID.
- PMOD 1 to 3 are not machined features; they mark where every board's Pmod host pin fields land.
- Board holes and slots are 3.4 mm, clearance for M3 standoffs. Plate fixings are 4.3 mm, clearance for M4. A slot serves two boards whose holes are too close together to drill separately.
- Pmod connector bodies overhang the front (lower) edge by 2.78 mm; keep it clear so a peripheral can plug in.
- Boards stand on M3 x 8 mm standoffs, plate face to board underside.

NOTES (continued)

- Cut file: tt-generic-mounting-plate.dxf. Sizes are finished sizes.
- Fit the plate to the chassis before the board: a board overhangs the plate fixings. Thinnest web between features is 1.59 mm, between A1 and E1.

SOURCES

- Derived from: [tinytapeout/boards.py](https://github.com/TinyTapeout/tt-demo-pcb/blob/521108f4abad8e7a57e517b170d3d62ceb06667f/case/tt06_demo_base.scad) - Board outlines, mounting holes and Pmod host positions for every Tiny Tapeout demo board revision, extracted from the upstream KiCad files.
- Standoff height: https://github.com/TinyTapeout/tt-demo-pcb/blob/521108f4abad8e7a57e517b170d3d62ceb06667f/case/tt06_demo_base.scad - Tiny Tapeout's own printed base: Height = 8 with a 1.6 mm PCB let into it, so its studs stand the board 6.4 mm up. The plate's 8 mm is not less.
- Pmod host pitch: <https://mith.ro/pmod-spec/> - Diligent mandate .90 in (22.86 mm) between adjacent host ports on a board edge, which is why one plate can serve every revision.

BOARD MOUNTING HOLES

ID	X mm	Y mm	DIA/WIDTH mm	LENGTH mm	USED BY
A1	23.45	81.78	3.40	-	DB mpw:MT3, DB ETR v3.2:MT1
A2	23.51	8.29	3.40	-	DB mpw:MT1
A3	120.51	8.29	3.40	-	DB mpw:MT2
A4	120.51	81.78	3.40	-	DB mpw:MT4
B1	111.81	8.47	3.40	-	DB 4+:MT2
B2	111.81	81.97	3.40	-	DB 4+:MT4
C1	106.75	13.22	3.40	-	DB 06+:MT2
C2	106.75	79.72	3.40	-	DB 06+:MT4
D1	96.40	14.27	3.40	-	DB ETR v3.2:MT2, DB ETR v3.3:MT2
E1	23.40	86.77	3.40	-	DB ETR v3.3:MT1
S1	14.24 / 14.80	8.72 / 8.47	3.40	4.01	DB 4+:MT1, DB 06+:MT1
S2	14.24 / 14.80	79.72 / 81.97	3.40	5.72	DB 4+:MT3, DB 06+:MT3

PLATE FIXING HOLES

ID	X mm	Y mm	DIA mm
P1	5.37	22.00	4.30
P2	5.37	74.00	4.30
P3	129.63	22.00	4.30
P4	129.63	74.00	4.30
P5	42.00	95.89	4.30
P6	93.00	95.89	4.30

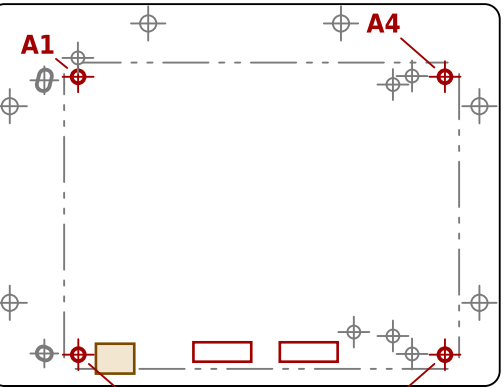
LEGEND

- Plate edge
- Hole or slot for a demo board fastener
- Plate fixing, into the chassis
- Slot axis
- Pmod host pin field, where a board's hosts land
- Pmod connector body, overhanging the front edge
- USB-C connector, one position per board version
- Dimension, extension and leader

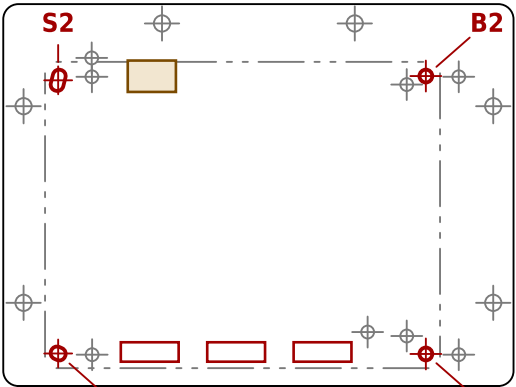
TT GENERIC MOUNTING PLATE

Accepts every demo board revision, Pmod hosts fixed in place

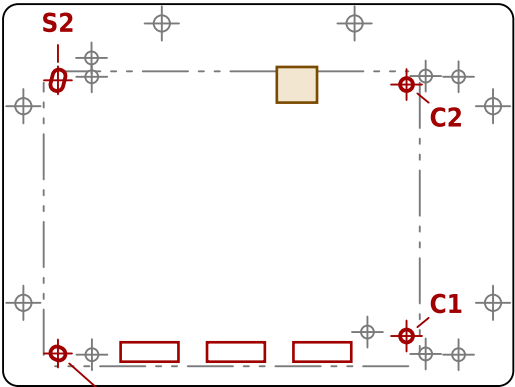
DRAWN generated		VERSION v0.0-352-g967eb38		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 2		DRAWING NO TT-MP-PLATE				REV A
MATERIAL 3 mm acrylic or 1.6 mm FR4			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 allow for cutter kerf				



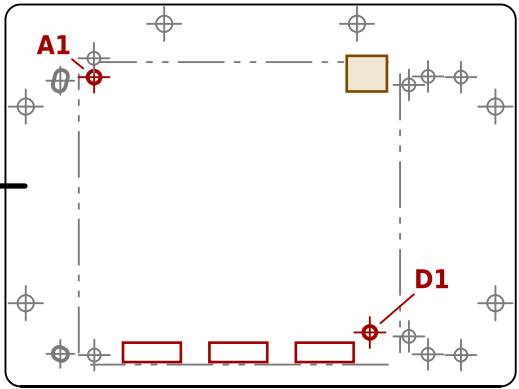
DB mpw
TT02, TT03
board rev tt123-v2.2.5, tt123-v2.2.6



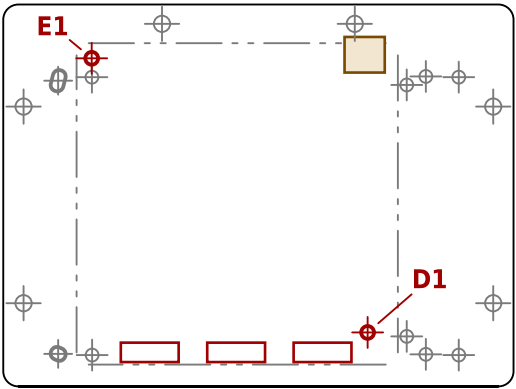
DB 4+
TT03p5, TT04, TT05
board rev v1.2.1, v1.2.2, v1.2.3



DB 06+
TT06, TT07, TT08
board rev v2.0.1, v2.1.0, v2.1.2



DB ETR v3.2
TT09, TTSKY25a, TTSKY25b, TTGF0p2
board rev v3.2



DB ETR v3.3
no shipped shuttle yet
board rev v3.3

NOTES

- Hole IDs are those of drawing TT-MP-PLATE, which governs every dimension.
- Revisions in a group share mounting holes and Pmod host positions exactly; they may differ elsewhere. v2.1.2's USB-C, for instance, is 0.9 mm from v2.0.1's.
- dX and dY place the board: add them to a coordinate in the board's own frame to get a plate coordinate.

BOARD PLACEMENT ON THE PLATE

BOARD	SHUTTLES	dX	dY	PMODS	HOLES USED
DB mpw	TT02, TT03	19.76	4.54	2, 3	A1, A2, A3, A4
DB 4+	TT03p5, TT04, TT05	11.05	4.72	1, 2, 3	B1, B2, S1, S2
DB 06+	TT06, TT07, TT08	10.74	5.22	1, 2, 3	C1, C2, S1, S2
DB ETR v3.2	TT09, TTSKY25a, TTSKY25b, TTGF0p2	19.40	5.77	1, 2, 3	A1, D1
DB ETR v3.3	not yet shipped	19.40	5.77	1, 2, 3	D1, E1

USB-C CONNECTOR POSITION ON THE PLATE

BOARD	X EXTENT mm	Y EXTENT mm
DB mpw	28.17 to 38.31	3.29 to 11.19
DB 4+	32.94 to 45.67	77.78 to 86.08
DB 06+	72.42 to 83.06	74.95 to 84.38
DB ETR v3.2	90.29 to 100.93	78.00 to 87.42
DB ETR v3.3	90.29 to 100.93	83.00 to 92.42

LEGEND

- Plate edge
- Hole or slot this revision uses
- Holes it does not use, and its board outline
- USB-C connector on this revision

TT MOUNTING PLATE FITTING GUIDE

Which holes each demo board revision uses

DRAWN generated		VERSION v0.0-193-gd9e6630		UNITS mm		SCALE 1:2	
SIZE A3	SHEET 2 OF 2		DRAWING NO TT-MP-FIT				REV A
MATERIAL - not a made part			GENERAL TOLERANCE reference only - TT-MP-PLATE governs every dimension				

1

2

3

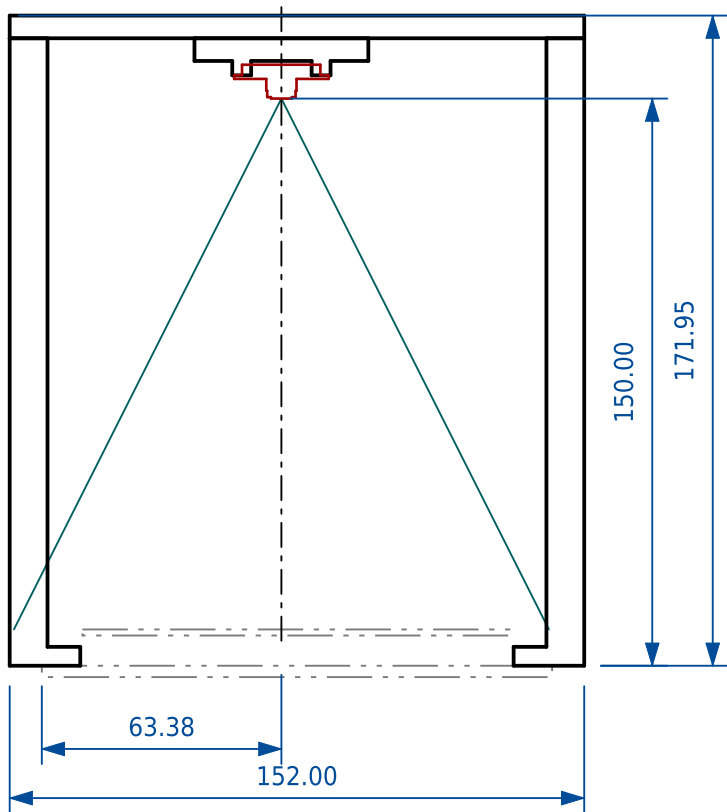
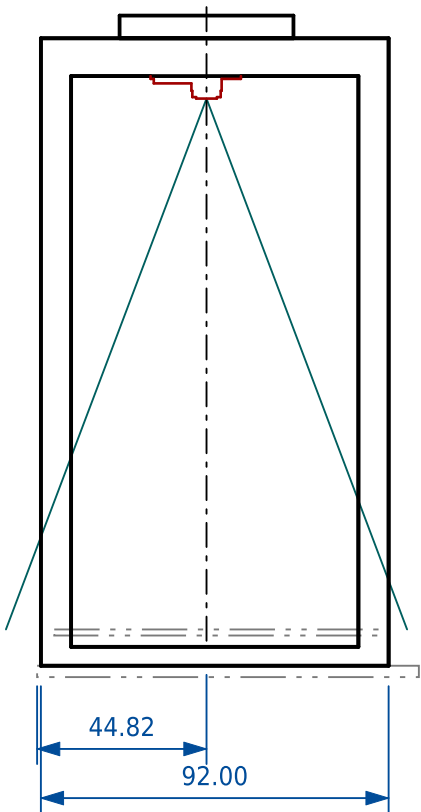
4

5

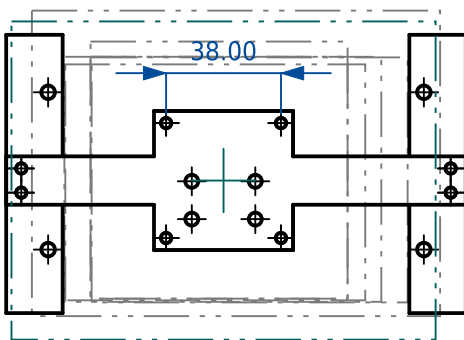
6

END ELEVATION

FRONT ELEVATION



PLAN, 1:2.5, NOT IN PROJECTION



NOTES

1. First angle, in TT-MP-PLATE's coordinates, Z up from the plate's TOP face. The right side frame is the left one mirrored.
2. FOR: the lens over RPICAM-OVER-PLATE's frame A, at (63.38, 44.82), high enough for the 65 deg lens, 53.50 x 41.41 deg, to take in every revision's whole board: 139.17 over the highest board face, 9.60 up, so 148.77. Set at 150.00, 1.23 over for the print, in a mode reading the whole sensor.
3. THE SAME BEAM AND CARRIER as TT-MP-CAM120, only 67.00 mm higher: only the side frames change with the lens.
4. TURN THE CAMERA RIGHT: ASSUMED that the picture's long side runs along the camera board's 25 mm width, which nobody publishes; it has to run along X, and a quarter turn out the picture is 17.1 mm short at each end. If it is, turn the carrier a quarter: its fixings are square about the lens.

NOTES (continued)

5. FOCUS: declared "Approx 1 m to infinity"; at 140.4 the boards are OUT OF FOCUS, a point about 21 pixels, 1.1 mm on the board (RPICAM-OVER-PLATE).
6. MAKE the parts from the STEP files, tt-camera-holder-65-side-left and -right and the shared -beam and -carrier; this sheet is the assembly. Holes: M4 4.5, M3 3.4, M2 2.4, M2 nut pockets 4.3 AF. Ream the side frames' holes, which print lying down. Opaque PETG, four perimeters, matte black if it can be: the side frames show in the picture's margin. The camera's holes are 2.0 +/-0.2: open a tight one with a 2.0 drill.

NOTES (continued)

7. ACCESS: front and back open over every Pmod host and USB-C, the side frames windows over the unfitted side Pmods J12-J14; buttons, not in the board data, are pressed from above, under a beam 166 mm up.
8. FIXING: the feet share the plate's own M4 side fixings, so no new hole; on a chassis the M4s are 5 mm longer than the plate's own. Plug the FFC in first -- its latch is out of reach under the carrier -- and take it up and back over the beam.

SOURCES

1. Where the camera goes: raspberry_pi_camera/optics.py - The plate subject's frame A and the 65 deg lens: RPICAM-OVER-PLATE.

PARTS, PRINTED

PART	QTY	X x Y x Z mm	PRINT
Side frame	2	18.7 x 92.0 x 165.9	Print on its outer face
Beam	1	152.0 x 46.0 x 6.0	Print flat, on its top face
Camera carrier	1	46.0 x 46.0 x 9.8	Print on its top face, bosses up

FASTENERS

QTY	SCREW	WITH	JOINS
4	M4 x 16 button head, ISO 7380	washer and nut	Foot to plate
4	M3 x 20 socket head, ISO 4762	and nut	Beam to the side frames' rails
4	M3 x 16 socket head, ISO 4762	and nut	Carrier to the beam's pad
4	M2 x 10 socket head, ISO 4762	and nut	Camera to the carrier

HEIGHTS ABOVE THE PLATE FACE, mm

	Z
Board faces	9.56 to 9.60
Lens face, least	148.77
Lens face, as built	150.00
Camera board	155.20 to 156.15
Carrier	159.95 to 165.95
Beam	165.95 to 171.95

LEGEND

- Printed part
- Camera Module, bought
- The plate and every revision's board, adjacent
- The 65 deg lens's field of view, to its picture's edge
- Dimension, extension and leader
- Frame A of RPICAM-OVER-PLATE, on the board plane

SOURCES (continued)

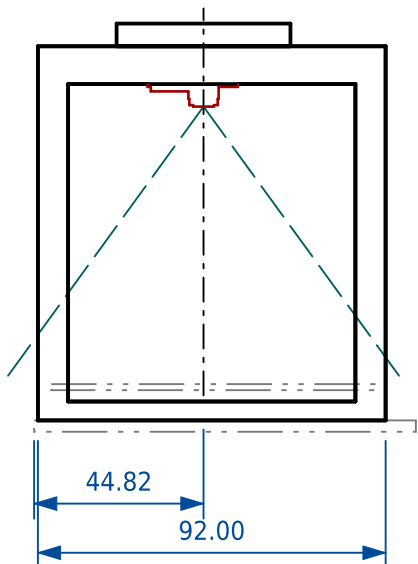
2. What it stands on: tinytapeout/mounting_plate/plate.py - Outline, side fixings, standoff; TT-MP-PLATE.
3. What it carries: raspberry_pi_camera/v1.py - The Camera Module v1.3: Gert van Loo's hand-measured sheet of 21 May 2013, checked against Raspberry Pi Spy's calipers.

CAMERA HOLDER, 65 DEG LENS

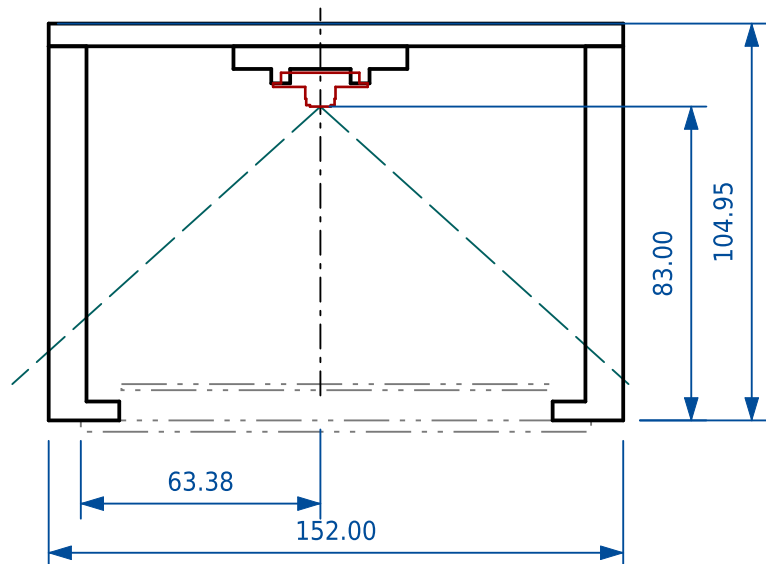
Holds a Camera Module over every demo board revision

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-352-g967eb38	mm	1:2, PLAN 1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-MP-CAM65	A
MATERIAL	GENERAL TOLERANCE	PROJECTION	
PETG, printed, opaque	print +/-0.2, Z DERIVED		

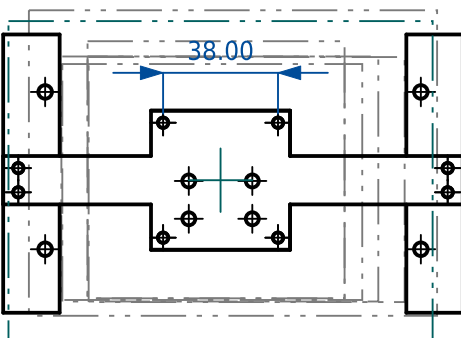
END ELEVATION



FRONT ELEVATION



PLAN, 1:2.5, NOT IN PROJECTION



NOTES

1. First angle, in TT-MP-PLATE's coordinates, Z up from the plate's TOP face. The right side frame is the left one mirrored.
2. FOR: the lens over RPICAM-OVER-PLATE's frame A, at (63.38, 44.82), high enough for the 120 deg lens, 96.00 x 72.00 deg, to take in every revision's whole board: 72.40 over the highest board face, 9.60 up, so 82.00. Set at 83.00, 1.00 over for the print, in a mode reading the whole sensor.
3. THE SAME BEAM AND CARRIER as TT-MP-CAM65, only 67.00 mm lower: only the side frames change with the lens.
4. CAMERA: a v1.3's board with a 120 deg lens ASSUMED on it, face 5.20 off the board; Arducam B006604 itself is a Pi Zero spy camera this carrier does not take. A taller lens lowers the face by the difference, and 1.00 is all there is: measure the lens first.

NOTES (continued)

5. TURN THE CAMERA RIGHT: ASSUMED that the picture's long side runs along the camera board's 25 mm width, which nobody publishes; it has to run along X, and a quarter turn out the picture is 16.8 mm short at each end. If it is, turn the carrier a quarter: its fixings are square about the lens.
6. FOCUS: declared "Focus Distance 1 m to infinity"; at 73.4 the boards are OUT OF FOCUS, a point about 18 pixels, 0.8 mm on the board (RPICAM-OVER-PLATE).
7. MAKE the parts from the STEP files, tt-camera-holder-120-side-left and -right and the shared -beam and -carrier; this sheet is the assembly. Holes: M4 4.5, M3 3.4, M2 2.4, M2 nut pockets 4.3 AF. Ream the side frames' holes, which print lying down. Opaque PETG, four perimeters, matte black if it can be: the side frames show in the picture's margin. The camera's holes are 2.0 +/-0.2: open a tight one with a 2.0 drill.
8. ACCESS: front and back open over every Pmod host and USB-C, the side frames windows over the unfitted side Pmods J12-J14; buttons, not in the board data, are pressed from above, under a beam 99 mm up.

NOTES (continued)

9. FIXING: the feet share the plate's own M4 side fixings, so no new hole; on a chassis the M4s are 5 mm longer than the plate's own. Plug the FFC in first -- its latch is out of reach under the carrier -- and take it up and back over the beam.

SOURCES

1. Where the camera goes: raspberry_pi_camera/optics.py - The plate subject's frame A and the 120 deg lens: RPICAM-OVER-PLATE.
2. What it stands on: tinytapeout/mounting_plate/plate.py - Outline, side fixings, standoff; TT-MP-PLATE.
3. What it carries: raspberry_pi_camera/v1.py - The Camera Module v1.3: Gert van Loo's hand-measured sheet of 21 May 2013, checked against Raspberry Pi Spy's calipers.

PARTS, PRINTED

PART	QTY	X x Y x Z mm	PRINT
Side frame	2	18.7 x 92.0 x 99.0	Print on its outer face
Beam	1	152.0 x 46.0 x 6.0	Print flat, on its top face
Camera carrier	1	46.0 x 46.0 x 9.8	Print on its top face, bosses up

FASTENERS

QTY	SCREW	WITH	JOINS
4	M4 x 16 button head, ISO 7380	washer and nut	Foot to plate
4	M3 x 20 socket head, ISO 4762	and nut	Beam to the side frames' rails
4	M3 x 16 socket head, ISO 4762	and nut	Carrier to the beam's pad
4	M2 x 10 socket head, ISO 4762	and nut	Camera to the carrier

HEIGHTS ABOVE THE PLATE FACE, mm

	Z
Board faces	9.56 to 9.60
Lens face, least	82.00
Lens face, as built	83.00
Camera board	88.20 to 89.15
Carrier	92.95 to 98.95
Beam	98.95 to 104.95

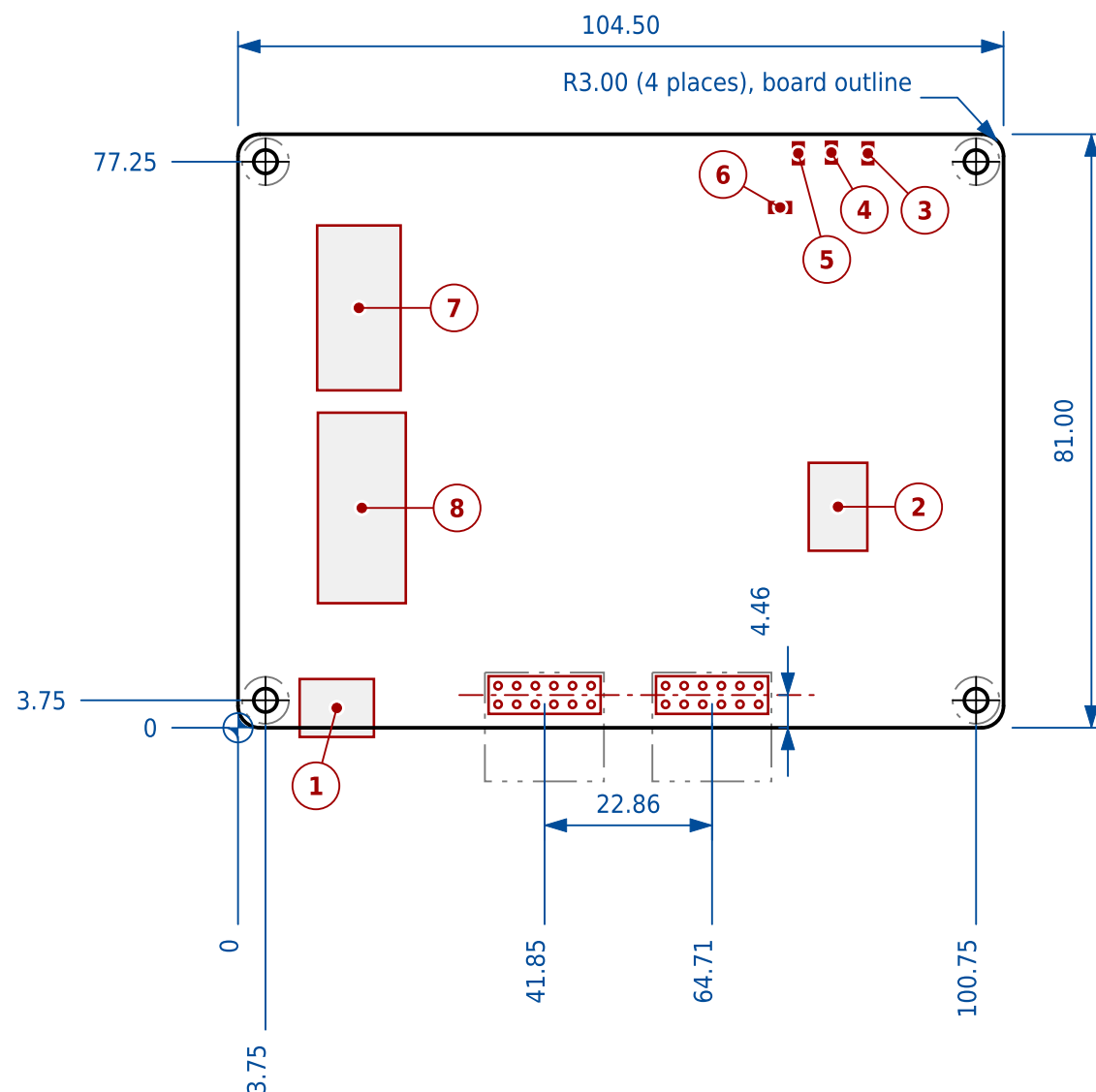
LEGEND

- Printed part
- Camera Module, bought
- The plate and every revision's board, adjacent
- The 120 deg lens's field of view, to its picture's edge
- Dimension, extension and leader
- Frame A of RPICAM-OVER-PLATE, on the board plane

CAMERA HOLDER, 120 DEG LENS

Holds a Camera Module over every demo board revision

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-352-g967eb38	mm	1:2, PLAN 1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-MP-CAM120	A
MATERIAL	GENERAL TOLERANCE	PROJECTION	
PETG, printed, opaque	print +/-0.2, Z DERIVED		



NOTES

1. Viewed from the component side.
2. Ships in TT02 Dev Kit and TT03 Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. Named "TinyTapeout 1,2,3 Demo Board" in KiCad, but no PCB served TT01, which was a bare-die trial run.
5. The revisions covered are mechanically identical; only the electrical design differs.
6. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-PLATE, registers against it.
7. Assembled envelope, connector overhang included: 104.50 x 82.25 mm.

SOURCES

1. KiCad board file:
https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb @ 303509a - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.5, dated 2023-11-13
2. KiCad board file:
https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb @ 3d721a6 - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.6, dated 2024-02-14
3. Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT02, TT03

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
PMOD A	J3	bottom	41.850	4.465	48.200	5.735
PMOD B	J9	bottom	64.710	4.465	71.060	5.735

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	8.41 to 18.55	-1.25 to 6.65
2	7-segment display	77.91 to 85.91	24.16 to 36.16
3	LED D1	85.25 to 86.72	76.92 to 79.88
4	LED D2	80.27 to 81.73	77.02 to 79.98
5	LED D3	75.75 to 77.22	76.92 to 79.88
6	LED D4	72.52 to 75.48	70.27 to 71.73
7	8-way input DIP switch	10.80 to 22.20	46.05 to 68.55
8	9-way DIP switch SW2	10.90 to 22.90	17.00 to 43.00
9	Side Pmod, first position - not on this board	-	-
10	Side Pmod, second position - not on this board	-	-
11	Side Pmod, third position - not on this board	-	-

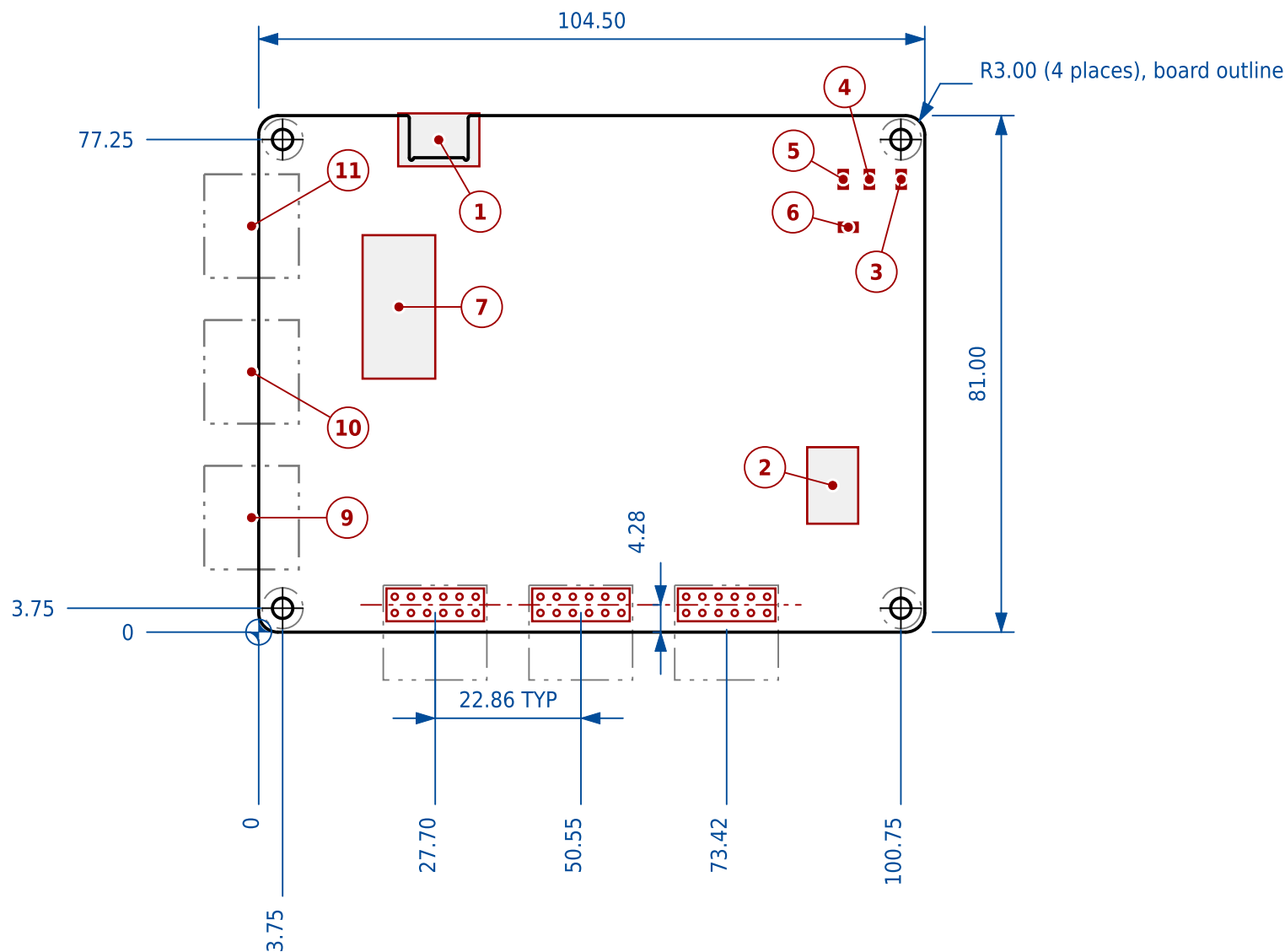
LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB MPW V2.2.5 / DB MPW V2.2.6

mpw-mb1 rev 2.2.5 and 2.2.6

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-193-gd9e6630	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-TT123	A
MATERIAL	GENERAL TOLERANCE		
PCB, 1.6 nominal	edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08		



NOTES

1. Viewed from the component side.
2. Ships in TT03p5 ASIC IC kit, TT04 Dev Kit and TT05 Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. TT03p5's board is recorded as rev 1.2.1 but unconfirmed: the TT03p5 render is labelled v1.1.2.
5. v1.2.2c is Tiny Tapeout's name for rev 1.2.3, an assembly variant of 1.2.2.
6. The revisions covered are mechanically identical; only the electrical design differs.
7. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-PLATE, registers against it.
8. Assembled envelope, connector overhang included: 104.50 x 81.35 mm.

NOTES (continued)

9. The notch in the upper edge is a recess for the USB-C shell.

SOURCES

1. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ 8aad3f8 - title block: TinyTapeout 4+ Demoboard rev 1.2.1, dated 2024-04-12
2. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
3. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
4. Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT03p5, TT04, TT05

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	27.695	4.275	34.045	5.545
BIDIR	J5	bottom	50.555	4.275	56.905	5.545
OUTPUT	J6	bottom	73.415	4.275	79.765	5.545

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	21.88 to 34.62	73.05 to 81.35
2	7-segment display	86.05 to 94.05	17.00 to 29.00
3	LED D1	100.07 to 101.53	69.52 to 72.48
4	LED D2	95.07 to 96.53	69.52 to 72.48
5	LED D3	90.97 to 92.43	69.52 to 72.48
6	LED D4	91.02 to 93.98	62.77 to 64.23
7	8-way input DIP switch	16.30 to 27.70	39.75 to 62.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-8.55 to 6.30	9.83 to 26.08
10	Side Pmod J13 (not fitted)	-8.55 to 6.30	32.69 to 48.94
11	Side Pmod J14 (not fitted)	-8.55 to 6.30	55.55 to 71.80

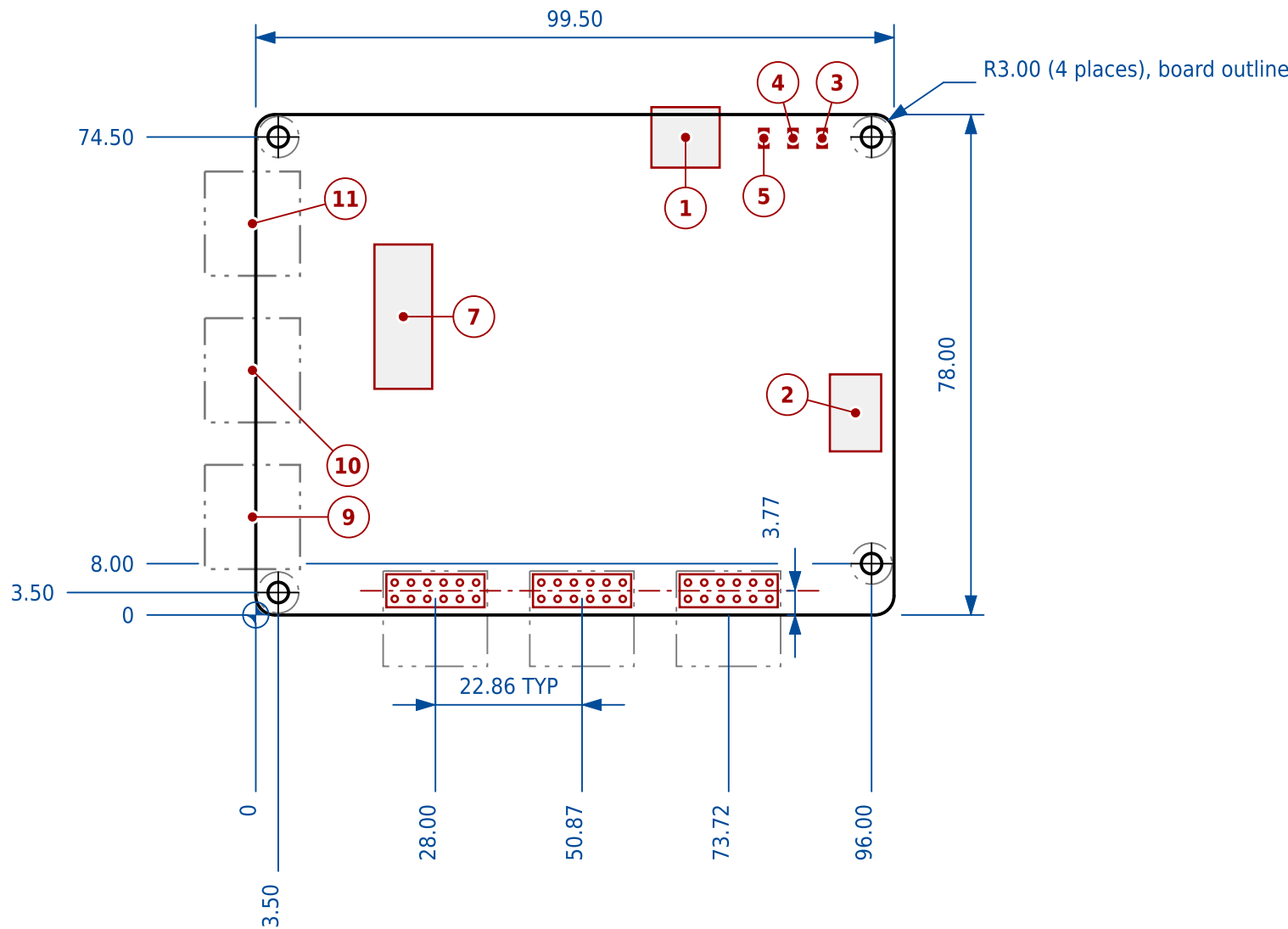
LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB 4+ V1.2.1 / DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.1, 1.2.2 and 1.2.3

DRAWN generated		VERSION v0.0-193-gd9e6630		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO TT-DB-V121				REV A
MATERIAL PCB, 1.6 nominal			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				



NOTES

- Viewed from the component side.
- Ships in TT06 Dev Kit and TT07 Dev Kit.
- Hole IDs are the KiCad reference designators.
- The revisions covered are mechanically identical; only the electrical design differs.
- Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-PLATE, registers against it.
- Assembled envelope, connector overhang included: 99.50 x 79.15 mm.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ 292760e1f - title block: TinyTapeout 06+ Demoboard rev 2.0.1, dated 2024-09-05
- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ a799acb38 - title block: TinyTapeout 06+ Demoboard rev 2.1.0, dated 2024-12-10
- Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT06, TT07

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	3.20	6.40
MT2	96.00	8.00	3.20	6.40
MT3	3.50	74.50	3.20	6.40
MT4	96.00	74.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	28.005	3.775	34.355	5.045
BIDIR	J5	bottom	50.865	3.775	57.215	5.045
OUTPUT	J6	bottom	73.725	3.775	80.075	5.045

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	61.68 to 72.32	69.73 to 79.15
2	7-segment display	89.50 to 97.50	25.50 to 37.50
3	LED D1	87.57 to 89.03	72.82 to 75.78
4	LED D2	83.02 to 84.48	72.82 to 75.78
5	LED D3	78.47 to 79.93	72.82 to 75.78
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	18.50 to 27.50	35.25 to 57.75
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-7.95 to 6.90	7.15 to 23.40
10	Side Pmod J13 (not fitted)	-7.95 to 6.90	30.01 to 46.26
11	Side Pmod J14 (not fitted)	-7.95 to 6.90	52.87 to 69.12

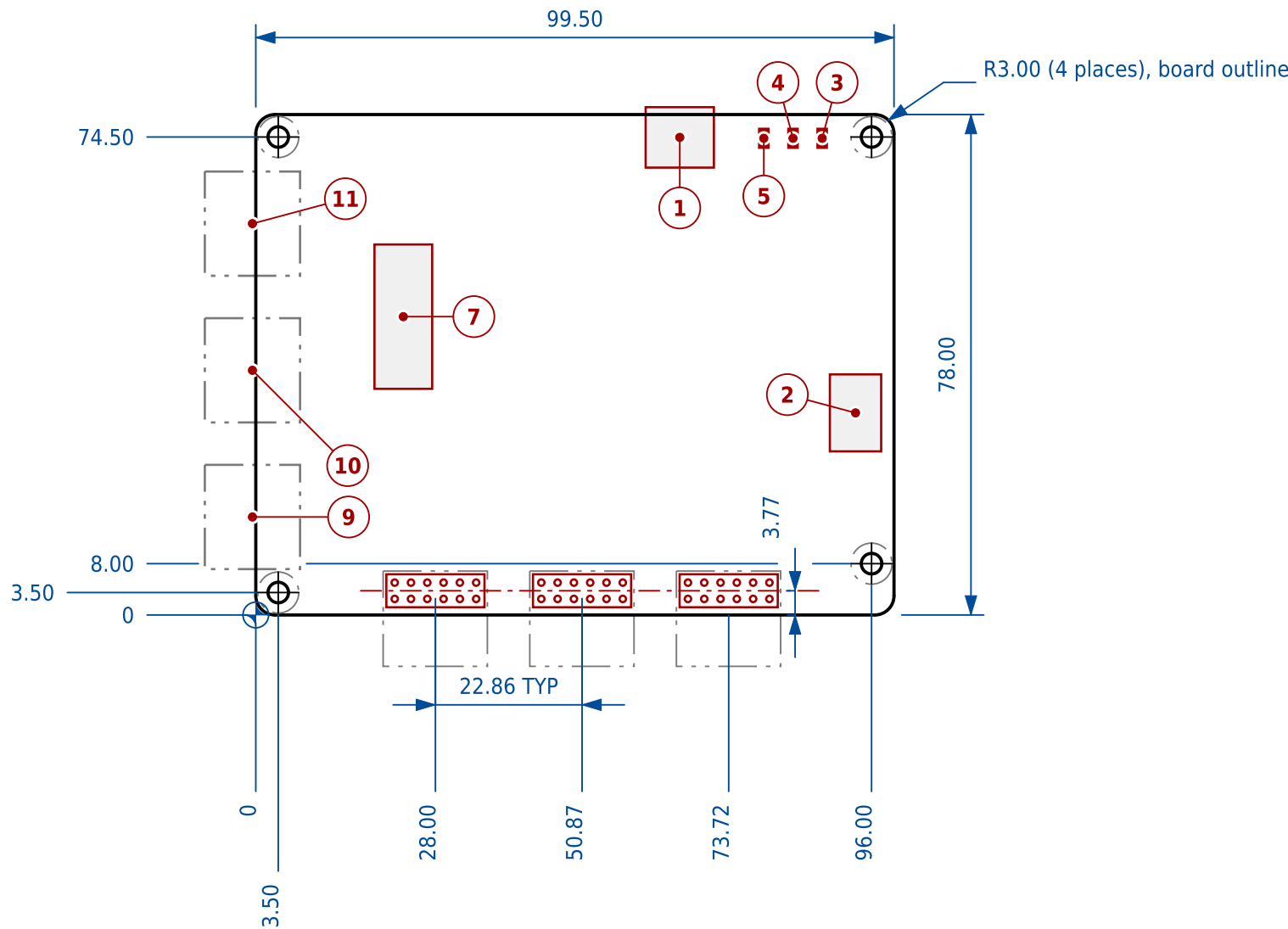
LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB 06+ V2.0.1 / DB 06+ V2.1.0

tinytapeout-demo rev 2.0.1 and 2.1.0

DRAWN generated		VERSION v0.0-193-gd9e6630		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO TT-DB-V201				REV A
MATERIAL PCB, 1.6 nominal			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				



NOTES

- Viewed from the component side.
- Ships in TT08 Dev Kit and TT08 Dev Kit - CoB edition.
- Hole IDs are the KiCad reference designators.
- Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-PLATE, registers against it.
- Assembled envelope, connector overhang included: 99.50 x 79.15 mm.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb>
tinytapeout-demo.kicad_pcb @ 028a51b1e - title block:
TinyTapeout 06+ Demoboard rev 2.1.2, dated 2025-02-11
- Shuttle mapping: <https://mith.ro/tt-demoboards/> - row "DB 06+ v2.1.2", Used by: TT08.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	3.20	6.40
MT2	96.00	8.00	3.20	6.40
MT3	3.50	74.50	3.20	6.40
MT4	96.00	74.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	28.005	3.775	34.355	5.045
BIDIR	J5	bottom	50.865	3.775	57.215	5.045
OUTPUT	J6	bottom	73.725	3.775	80.075	5.045

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	60.78 to 71.42	69.73 to 79.15
2	7-segment display	89.50 to 97.50	25.50 to 37.50
3	LED D1	87.57 to 89.03	72.82 to 75.78
4	LED D2	83.02 to 84.48	72.82 to 75.78
5	LED D3	78.47 to 79.93	72.82 to 75.78
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	18.50 to 27.50	35.25 to 57.75
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-7.95 to 6.90	7.15 to 23.40
10	Side Pmod J13 (not fitted)	-7.95 to 6.90	30.01 to 46.26
11	Side Pmod J14 (not fitted)	-7.95 to 6.90	52.87 to 69.12

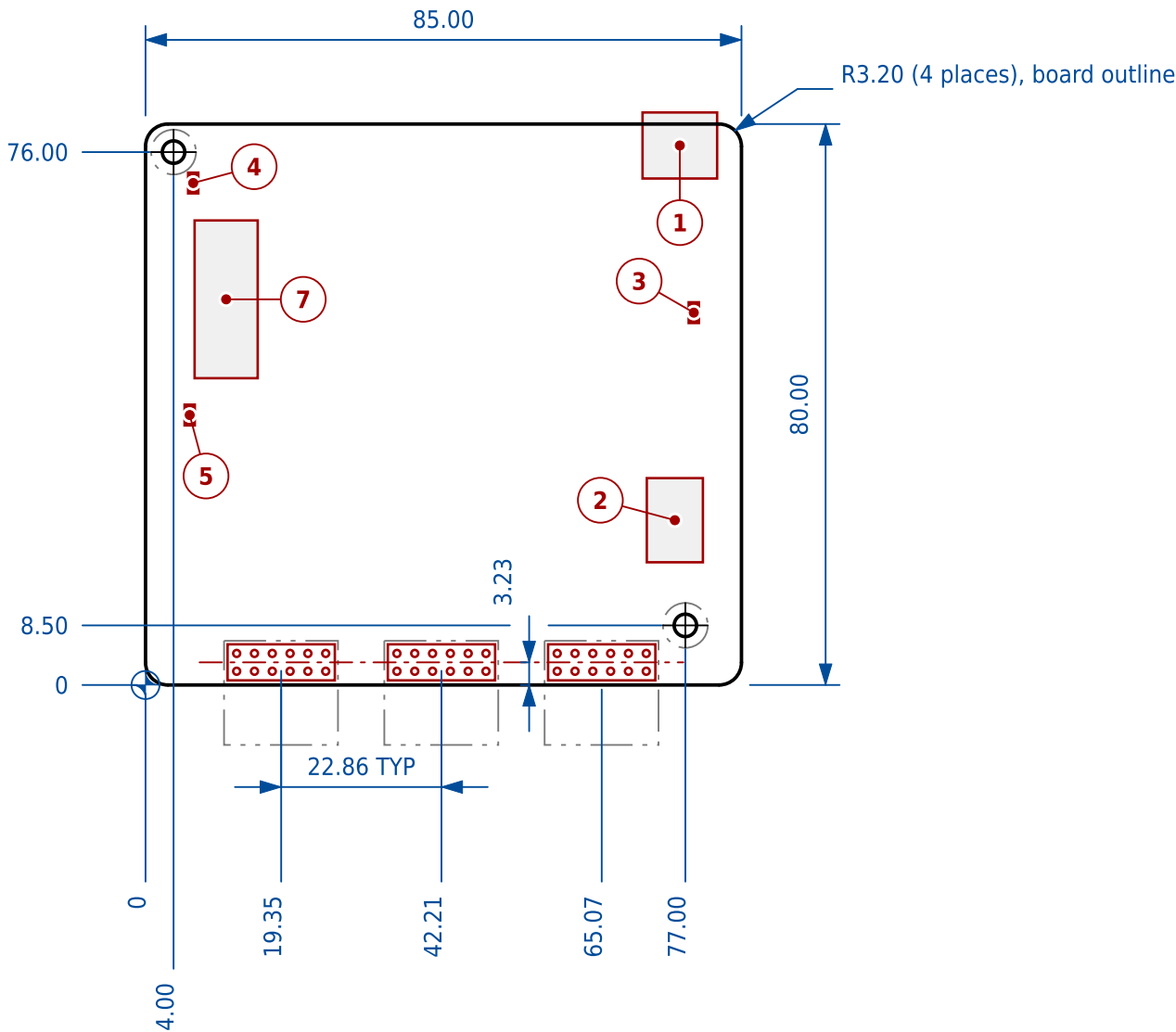
LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB 06+ V2.1.2

tinytapeout-demo rev 2.1.2

DRAWN generated		VERSION v0.0-193-gd9e6630	UNITS mm	SCALE 1:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO TT-DB-V212			REV A
MATERIAL PCB, 1.6 nominal		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			



NOTES

1. Viewed from the component side.
2. Ships in TT09 Dev Kit, TTSKY25a Dev Kit, TTSKY25b Dev Kit, TTGF0p2 Dev Kit - P2 edition and FPGA Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. "DB ETR" is the board's ID in Tiny Tapeout's board revision spreadsheet; the KiCad title block does not carry it.
5. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-PLATE, registers against it.
6. Assembled envelope, connector overhang included: 85.00 x 81.65 mm.

SOURCES

1. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ d830790ca - title block: Tiny Tapeout Demoboard v3 rev 3.2, dated 2025-11-12
2. Shuttle mapping: <https://mith.ro/tt-demoboards/> - row "DB ETR v3.2", Used by: TT09, TTSKY25a, TTSKY25b, TTGF0p2.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	4.00	76.00	3.20	6.40
MT2	77.00	8.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J11	bottom	19.350	3.230	25.700	4.500
BIDIR	J12	bottom	42.210	3.230	48.560	4.500
OUTPUT	J13	bottom	65.070	3.230	71.420	4.500

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	70.89 to 81.53	72.23 to 81.65
2	7-segment display	71.50 to 79.50	17.50 to 29.50
3	LED D3	77.47 to 78.93	51.62 to 54.58
4	LED D4	6.07 to 7.53	70.11 to 73.07
5	LED D5	5.57 to 7.03	37.02 to 39.98
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	7.00 to 16.00	43.75 to 66.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod, first position - not on this board	-	-
10	Side Pmod, second position - not on this board	-	-
11	Side Pmod, third position - not on this board	-	-

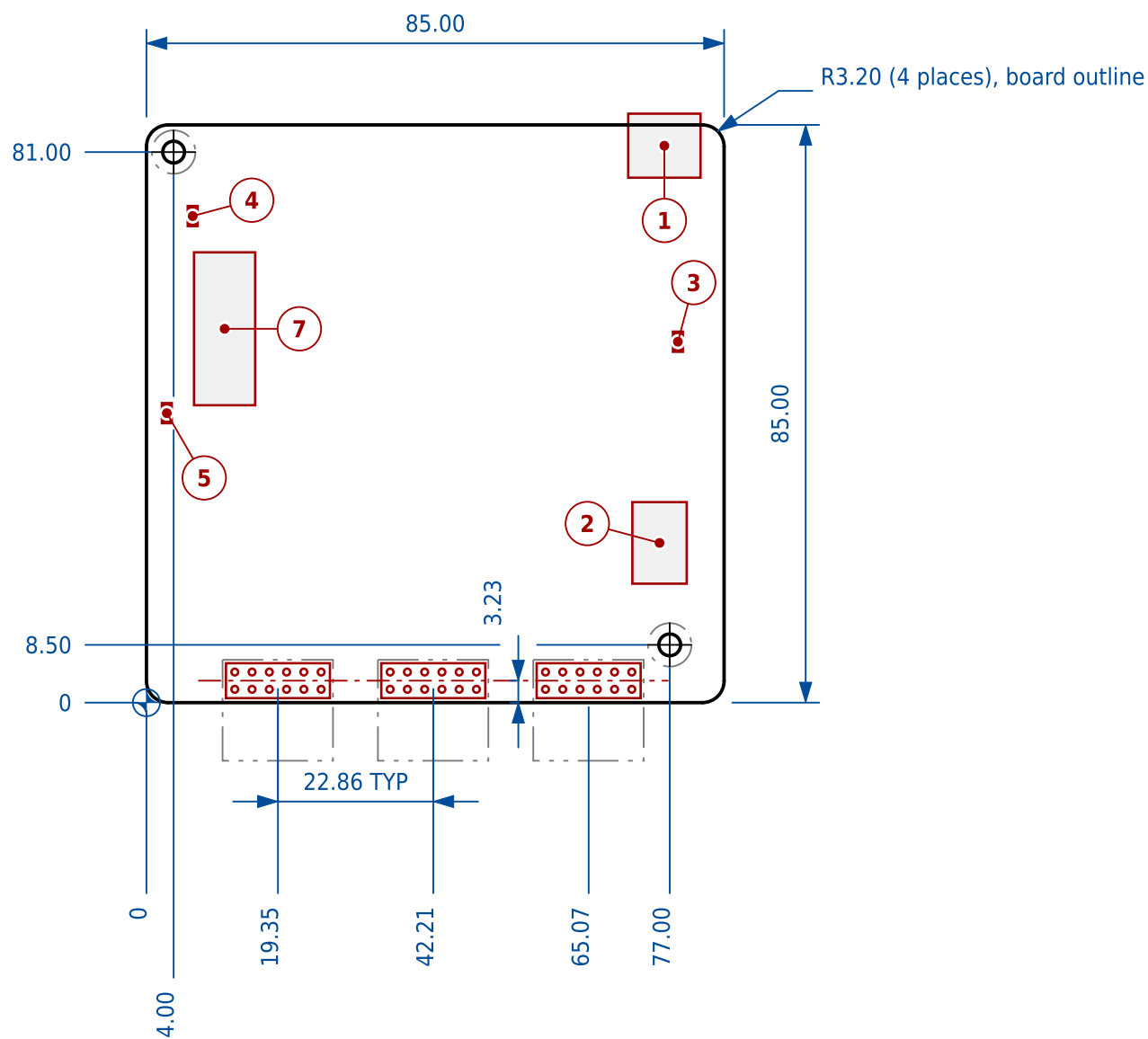
LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB ETR V3.2

tinytapeout-demo rev 3.2

DRAWN generated		VERSION v0.0-193-gd9e6630	UNITS mm	SCALE 1:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO TT-DB-V32			REV A
MATERIAL PCB, 1.6 nominal		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			



NOTES

- Viewed from the component side.
- Hole IDs are the KiCad reference designators.
- Not in Tiny Tapeout's board revision spreadsheet, so no shuttle or kit is known to have shipped on it; the name follows the v3.2 ID.
- Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-PLATE, registers against it.
- Assembled envelope, connector overhang included: 85.00 x 86.65 mm.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ ecb636ace - title block: Tiny Tapeout Demoboard v3 rev 3.3, dated 2026-06-23
- Shuttle mapping: <https://mith.ro/tt-demoboards/> - no row: this revision is not in the board revision spreadsheet, so no shuttle is recorded for it.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	4.00	81.00	3.20	6.40
MT2	77.00	8.50	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J11	bottom	19.350	3.230	25.700	4.500
BIDIR	J12	bottom	42.210	3.230	48.560	4.500
OUTPUT	J13	bottom	65.070	3.230	71.420	4.500

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	70.89 to 81.53	77.23 to 86.65
2	7-segment display	71.50 to 79.50	17.50 to 29.50
3	LED D3	77.47 to 78.93	51.62 to 54.58
4	LED D4	6.07 to 7.53	70.11 to 73.07
5	LED D5	2.27 to 3.73	41.11 to 44.07
6	Fourth indicator LED - not on this board	-	-
7	8-way input DIP switch	7.00 to 16.00	43.75 to 66.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod, first position - not on this board	-	-
10	Side Pmod, second position - not on this board	-	-
11	Side Pmod, third position - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB ETR V3.3

tinytapeout-demo rev 3.3

DRAWN generated		VERSION v0.0-193-gd9e6630		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO TT-DB-V33				REV A
MATERIAL PCB, 1.6 nominal			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				